

Agrivoltaic Farming

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What is Agrivoltaic Farming?

- Agrivoltaic farming is the practice of growing crops underneath solar panels.
- A farm full of crops grown under solar panels.



Advantages of Agrivoltaic farming

Land - efficiency

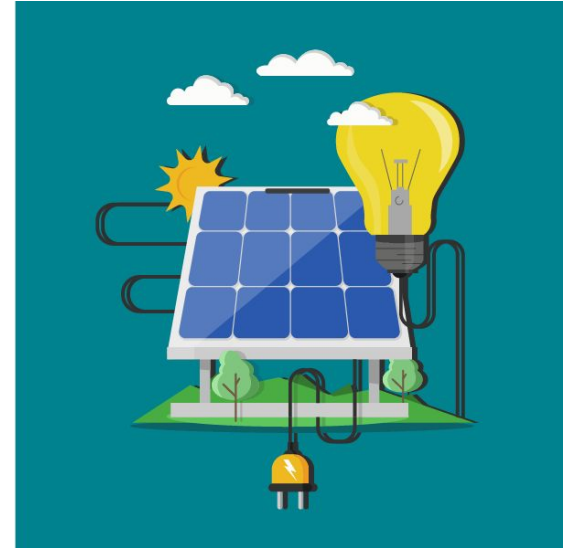
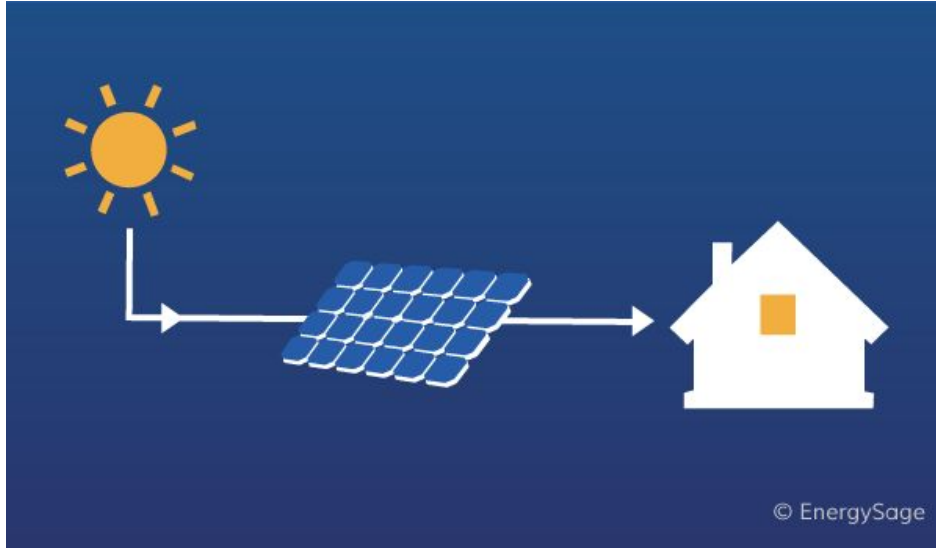


Land - efficiency



Agricultural + Photovoltaic = Agrivoltaic

More energy



Better Crops



Better Crops



Better Crops with less water?



Other methods of agrivoltaics:

1. Soil and vegetation can reduce erosion
 - a. No need for paved drainage systems
2. Grass and overgrown vegetation can be a major pain on solar sites. Sheep can help!
 - a. Eliminates the need for mowing
 - b. Provides dairy, meat, and wool
3. Encourages the natural habitat
 - a. Habitat for pollinators
 - b. Can attract not so nice wildlife such as rodents and insects



Unique challenges to agrivoltaics:

1. Panels may indeed to be installed higher to allow sunlight and farm equipment through.
2. Construction equipment can compact soil and make it harder to hold water and nutrients.
 - a. Vehicles can also kick dust onto panels and reduce their efficiency.
3. Farmers need access underneath panels. To do this, they may need to change their farming methods, use special equipment, and have electrical conductors properly covered. [2]



Unique challenges to agrivoltaics:

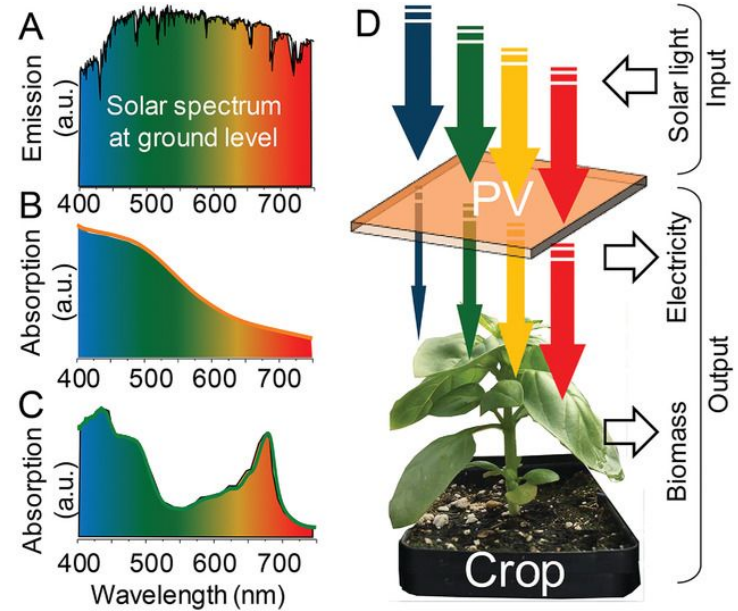
Agrivoltaic are very plant and site specific

“Tomatoes saw up to double yield with agrivoltaics, while wheat, cucumbers, potatoes and lettuce showed significant negative impacts and corn and grapes showed minimal impact.”

Agrivoltaic tomato growth was tested in Japan. Growth was 10% lower than conventional agriculture. The same test performed in Morocco showed no difference in growth compared to an unmodified greenhouse

Colder countries like Germany or Denmark have limited sunlight, so Panels may need to be spaced further apart. [3]

Plants and PV are both photovoltaic systems. How do you share the sunlight?



Plants can only use a certain amount of light before it gets to a “light saturation point”. Excess/harmful solar energy (e.g., UV) is dissipated protect the plants from light stress. Therefore, part of the solar energy absorbed in the blue and green portions of the electromagnetic spectrum is dissipated without contributing to photosynthesis. [4]

Temperature Analysis

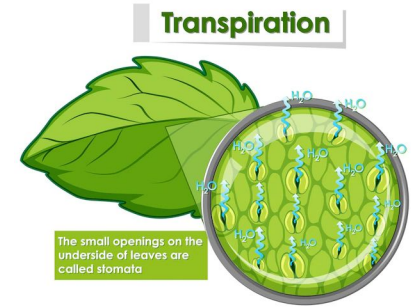
Solar arrays are less efficient when they're hot

Plants have a cooling effect on the panels due to transpiration.

One study in the UK observed cooling of up to 5.2 °C in the summer and 1.7 °C in the winter compared to control areas

How much more power does that give us?

In a 2019 study, solar panels were kept 9°C cooler by evaporation from crops. This increased their power production by 2%



TMY data: June 12th 3pm	Phoenix	UK
Irradiance	76.97 mW/cm^2	57.18 mW/cm^2
Dry bulb temp °C	43.3°C	25°C
Tcell	67.35°C	42.8°C
Power loss	14.82%	6.2%
Power loss (with plants)	13%	4.5%
Plant savings (20 MW system)	\$320 per day (in summer)	\$349 per day (in summer)



$$T_{Cell} = T_{Air} + \frac{NOCT - 20}{80} S$$

TRADE-OFFS:

Raising the height of PV modules can increase the cost of the solar installation. Due to the increased wind loading on higher structures, the length of steel foundational posts underground would likely need to be increased to accommodate the additional structural loading.

Solar can be installed in flood plains, but all electrical equipment will have to be installed above the projected level of flooding. Raising equipment could increase the cost of installation and may negatively impact the project economics. Also, the cost of insurance will be higher for PV systems in a flooding area. An area that will not be flooded may be better suited for PV installation.

If there is a desire to grow crops underneath and in between solar modules, smaller tractors or hand management are options. There is no one-size-fits-all solar design and developers should account for land and farming needs in the design process.

There is likely to be more of a collision risk to birds presented by infrastructure associated with solar PV developments, such as overhead power lines.

TRADE-OFFS:

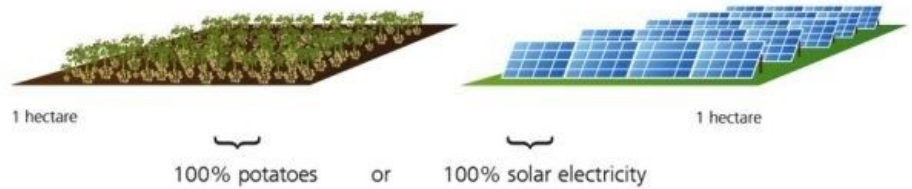


Currently, the main disadvantage of agrivoltaics is the high upfront cost required. Compared to conventional roof mounted PV, dual use solar systems require a larger, more complex mounting systems which can cost up to triple the cost of the installation of flat roof PVs. Further to this, deeper steel foundations are required for the PVs to provide stability and resilience in adverse weather conditions; the embodied carbon emissions associated with this are high.

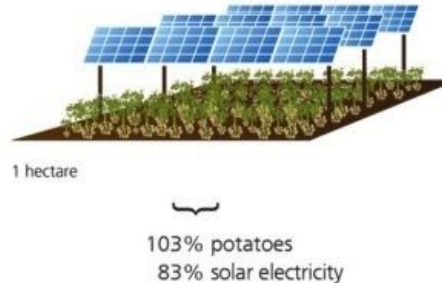
Performance:

- Improving the ecosystem by using less space for 2 economic friendly sources by increasing the electricity production and efficiency of growth in crops.
- Shade provided by the solar panels, help the plants maintain water levels. Essentially saving money on water.
- The crops keep the solar panels cooler than without the crops, making them more efficient.
- Solar panels: Too hot = Less efficient
- Chiltepin pepper tripled, and cherry tomato doubled in this environment.

Separate Land Use on 1 Hectare Cropland: 100% Potatoes or 100% Solar Electricity



Combined Land Use on 1 Hectare Cropland: 186% Land Use Efficiency



Commercial Issues:

- Finding good locations in terms of agriculture and that have enough sunlight for solar panels. Choosing the crop in this environment may be difficult.
- Mold and soil damage can be caused from the construction equipment.
- More expensive due to larger and more complex mounting systems for the panels, as they are often being irrigated because of the crops.
- Rodents and insects can become attracted to the moisture and the humidity, that can cause them to live near the crops and eat through the electrical wiring.
- Research is still being done, governments and private companies will steer away from this idea if no long term success like lack of financial gain.
- Possibly have to go through many loopholes because of regulations.

Summary

Preserving farmlands while expanding the renewable energy generation is an important aspect of meeting future energy needs as increasing densely populated areas hinder the renewable capacity expansion. Depending on the climatic conditions of a region, agro PV provides an ample number of opportunities in achieving a sustainable solution towards energy-food combination. APV adds value by enabling simultaneous production of energy and food, which correlates to economic benefits to farmers along with the potential to provide synergistic effects. Adopting shade-resistant crops in hot-arid climatic regions can enhance the yield while mitigating adverse effects of excessive temperatures due to high radiations. APV possibly lowers the cost of the energetic transition while at the same time does not consume any additional land.

References

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A photograph of a solar farm. In the foreground, there are rows of lush green plants, likely tomatoes, growing in a field. Above the plants, there are long, parallel rows of solar panels mounted on metal supports. The panels are tilted towards the sky. The background shows a clear blue sky and the continuation of the solar farm structure.

THANK YOU